C/OS-II

Supported Platform: ARM Cortex-M3, ARM Cortex-M4F, ARM ARM7TDMI, Atmel AVR

Kernel Size: 20K

MicroC/OS-II is a free for non-commercial use developed by Jean J. Labrosse for a two-part article in Embedded Systems Programming magazine, much like Andrew Tanenbaum who created Minix when writing a textbook. µC/OS-II has been ported to over 100 different platforms ranging from 8-bit to 64-bit microprocessor, microcontrollers and DSPs. This OS is also vetted by the US FAA for use in avionics.

RTLinux

Supported Platform: All hardware supported by Linux

Kernel Size: 125-256K

RTLinux runs along with a stock Linux system. The real-time kernel sits between the hardware and the actual Linux system. The RT kernel acts as a proxy and intercepts all the interrupts from the hardware. If an interrupt causes a real-time task to run, the real time kernel preempts Linux and lets the real time task run it's course. In a way, Linux itself runs as a task of the real time system. The real time tasks are loaded as kernel modules, and thus they're part of the kernel and run in kernel space. It is used to control robots, data acquisition systems, manufacturing plants, and other time-sensitive instruments and machines.

QNX

Supported Platform: Intel 8088, x86, MIPS, PowerPC, SH-4, ARM, StrongARM, XScale

Kernel Size: 12K

QNX has a long and colorful history. In it's most current avatar it's being called Blackberry 10 is running Blackberry's newest smartphones. The microkernel architecture offers immense advantage as the ability to finely tune the OS to your exact specification is immensely beneficial to manufacturers producing components in large volume, where even 1% reduction in memory can reduce the cost by millions of dollars.